AI Video Studio — Complete Project Structure

What the project is made of, the hardware it runs on, every API and tool it uses, and what each one does. Folder: `~/yt-automation-studio` · Private GitHub repo `RaviBhardawaj/YT-Automation` · Status as of 30 Oct 2026

> **In one line:** a local app on your Mac that researches a topic, writes the script, fact-checks it, gathers visuals, records the voice, edits the video with FFmpeg, checks quality and uploads to YouTube — on a daily schedule, per channel, using free services first (budget ₹0).

1. How one video is made (the pipeline)

Research → Script → Fact-check review → Visuals → Voice → Render → Quality check → Upload

Stage	What happens	Tools / APIs used
Topic	Daily schedule picks a topic per channel. AI News: today's most-searched story that ≥2 real outlets report (astrology excluded, no repeats within 3 days).	Google Trends RSS, Google News RSS, YouTube Data API
Research	Collects evidence: articles, Wikipedia, YouTube competitors, page text. Every number comes from a source.	Google News, Wikipedia + pageviews, YouTube search, Hacker News, RSS, Ollama web search
Script	AI writer produces scenes (narration, visuals, on-screen text), title, description, tags, chapters, thumbnail idea, fact list. Long videos are written chapter by chapter with length checks.	Free writer chain (section 4)
Review	A strict "editor" AI checks every claim against the evidence. If something is wrong: full rewrite once, then only the flagged lines are rewritten and re-checked (2 rounds). Still wrong → waits for your approval.	Gemini / free writer chain
Visuals	Each scene gets a picture or clip: stock media checked by a vision model for relevance, AI-drawn images, or original graphics. AI News: the photo-real presenter opens and closes the video.	Pixabay, Openverse, Wikimedia, Cloudflare FLUX, Hugging Face FLUX, local FLUX, procedural art
Voice	Narration per scene with timings for captions.	Kokoro (local), Piper (local), Sarvam (cloud, paid), Chatterbox voice clone (local)
Render	FFmpeg builds the video: motion on stills, captions, headlines, music, transitions, thumbnail, subtitles. Presenter scenes become talking clips when GPU allows.	FFmpeg, Pillow, LTX-2 talking clips, SadTalker (on the Mac)
Quality check	Resolution, fps, codecs, length, audio loudness, clipping, full decode, black frames, visual variety, editorial review, finance wording, viral checklist.	FFmpeg / ffprobe, built-in rules
Upload	Resumable upload, thumbnail, captions, category, "altered or synthetic content" disclosure for AI presenters. Then analytics are pulled daily and each channel learns from its results.	YouTube Data API v3, YouTube Analytics API (official OAuth)

2. Hardware and system

Part	What you have	What it means for the studio
Computer	Apple MacBook with **Apple M2** chip	Runs everything locally; its GPU (Metal/MPS) is used for local AI images and SadTalker.
Memory (RAM)	**16 GB**	Enough for voices, Whisper, 4-bit FLUX, SadTalker. Too small for big local video models (LTX-2, Wan) — those run on free cloud GPUs.
Storage	245 GB disk, **~12 GB free**	**Tight. Finished videos are ~25–125 MB each. The Storage page shows live memory/disk use and has an Uninstall button for each AI tool.**
Operating system	macOS 26.6.2	App runs as a LaunchAgent (starts with the Mac, restarts if it crashes).
Internet	iPhone hotspot (mobile data)	IPv6 is unreliable on it, so the app always uses IPv4. Big downloads need your OK.
Web dashboard	`http://127.0.0.1:8765`	Only reachable from this Mac. Sign-in required.

3. Code structure (folders and files)

About 140 Python files in `studio/`, 33 test files (359 automated tests).

Folder / file	What it does
`studio/web/app.py`	The dashboard (FastAPI + Jinja2 pages): channels, videos, approvals, settings, credentials, storage, Film Studio, Shorts clipper, Upload & Edit.
`studio/pipeline.py`	The production state machine (stages above). Every stage saves its output, so a crash or restart resumes where it stopped.
`studio/worker.py`, `tasks.py`, `scheduler.py`, `jobs.py`	Task queue with two worker lanes (media / general), retries and priorities; daily schedules per channel (missed runs run on wake); one entry point for every new video.
`studio/channels.py`	Many YouTube channels in one studio: niche, language, style, workflow, up to 4 daily times, voice, publishing rules.
`studio/ai/`	`router.py` picks the AI provider (free first) and validates every answer; `providers.py` talks to Ollama, Gemini, Cloudflare, Claude; `prompts.py`, `schema.py`; `videogen.py` (paid Gemini video, unused).
`studio/content/package.py`	Script, storyboard, metadata, fact-check review and the flagged-line fixer.
`studio/research/`	Evidence collectors (news, Wikipedia, YouTube, RSS), topic ranking, duplicate and copy detection.
`studio/media/`	`visuals.py` (stock + AI images), `vision.py` (is this picture on topic?), `tts.py` (voices), `voiceclone.py`, `captions.py`, `graphics.py`, `render.py` (FFmpeg), `thumbnail.py`, `qa.py`.

`studio/creative/`	Creative channels: `news.py` (AI News + presenter), `cartoon.py` and `workflow.py` (animated kids/comedy episodes), `characters.py`, `screenplay.py`, `film.py` (photo-real films), `music.py` (original music), `trends.py`, `viral.py`, `branding.py`.
`studio/film_studio/`	Photoreal Film Studio: casting, shot list, shot generation, `zerogpu.py` (Wan 2.2 motion clips), `talking.py` (LTX-2 lip-synced clips), `sadtalker.py` (talking presenter on the Mac), edit, QA, Shorts from films.
`studio/finance/`	NIFTY 50 Fibonacci channel: verified market data, pivot maths, charts, Hindi/English wording, lessons, SEBI-safe wording checks, compliance gate, hash-chained audit log.
`studio/editing/`	Upload & Edit and the AI Shorts clipper: local Whisper transcription, AI edit plan, FFmpeg editing engine.
`studio/youtube/`	`oauth.py` (Google sign-in, per channel), `upload.py`, `analytics.py`, `growth.py` (monetization progress).
`studio/social/`	Facebook Pages and Instagram publishing via Meta Graph API (ready, not connected).
`studio/storage.py`	Storage page: delete videos, clear working files, live Mac memory/CPU/disk use, uninstall optional AI tools.
`studio/settings.py`, `auth.py`, `plans.py`, `db.py`, `net.py`, `storage.py`, `learning.py`	Settings + Keychain secrets; accounts and sessions; plan limits; SQLite database; safe networking (IPv4, timeouts); disk clean-up; daily learning from each channel's results.
`data/` (not in Git)	Database, videos, voices, models, logs (`data/logs/studio.log`).
`tests/`, `docs/`, `install/`, `deploy/`	Automated tests; reports and guides; installer scripts; LaunchAgent setup.

4. AI writers (scripts and fact-checking)

The router tries these in order and moves on when one is used up. If all are used up, the video **waits 30 minutes and tries again** instead of failing.

Provider	Model	Cost / limit	Status today
Ollama (local, on the Mac)	qwen3.5:9b	Free, unlimited — but ~6 GB download	**No model downloaded**
Ollama Cloud	gemma4:31b	Free monthly allowance	**This month's limit reached**
Google Gemini API	Rotates 7 free models: gemini-3.5-flash, 3-flash-preview, 3.5-flash-lite, 3.1-flash-lite, flash-lite-latest, Gemma 4 31B, Gemma 4 26B	Free tier, each model has its own small daily limit	**Working — wrote today's videos**
Cloudflare Workers AI	Llama 3.3 70B	Free 10,000 "neurons" a day (shared with Cloudflare images)	**Working (backup)**
Claude API (Anthropic)	Claude Opus	Paid	Key saved; skipped while budget is ₹0

5. Pictures, video clips and talking presenter

Service / tool	What it does	Cost / limit	Status
Pixabay API	Free stock photos and videos	Free	**Key set**
Pexels API	Free stock photos and videos	Free	**No key yet**
Openverse, Wikimedia Commons	Openly licensed photos (with attribution)	Free, no key	**Working**
Cloudflare FLUX.2 klein	AI images (presenter, scenes, thumbnails)	Free daily neurons	**Key set**
Hugging Face FLUX	AI images (backup)	Free with token	**Token set**
Local FLUX.2 klein 4-bit (mflux)	AI images on the Mac's GPU, no internet needed	Free, unlimited	**Installed (`.venv-flux`)**
Vision check	Looks at each stock picture and scores whether it fits the scene	Free model	**Working**
Procedural art	Original graphics/text cards drawn by the app	Free	**Always available**
Wan 2.2 (HF ZeroGPU Space)	Turns a still into a short real-motion clip	Free GPU minutes, shared	**Few clips a day**
LTX-2 (HF ZeroGPU Space `alexnasa/ltx-2-TURBO`)	Most realistic talking presenter: lip-synced to our voice, 3-second pieces chained	Free GPU minutes: ~2 pieces a day	**Used up today; resets 1:56 PM**

SadTalker (on the Mac)	Talking face from one photo + our voice; runs on the Mac with no daily limit (Apache 2.0)	Free, unlimited; ~1 min of Mac time per 1 s of speech	**Installed (`~/sadtalker`, 3.1 GB)**
Gemini Omni Flash video	Text-to-video	Paid tier only	Not used

6. Voices, music and speech-to-text

Tool	What it does	Cost / licence	Status
Kokoro (local)	Natural Hindi/English narration; AI News uses the male voice `hm_omega`	Free, Apache 2.0	**Installed**
Piper (local)	Lightweight backup voices (Hindi Piper voices are non-commercial, so not used for Hindi)	Free, licence per voice	**Installed**
Sarvam AI Bulbul	Cloud Hindi voices ("ritu")	Paid per character	**Out of credits**
Chatterbox Multilingual	Your own cloned voice (with consent)	Free, MIT	**Installed (`.venv-voiceclone`)**
Indic Parler-TTS	Indian-language voices with style control	Free	**Installed (`.venv-parler`)**
faster-whisper	Speech-to-text for editing and Shorts clipping	Free, local	**Installed**
Music engine	Original background music generated by the app	Free, no samples	**Built in**

7. Platforms, research and data APIs

API	What it does	Cost / limit	Status
YouTube Data API v3 (OAuth)	Upload, thumbnails, captions, channel info; competitor research	Free, 10,000 units/day	**Connected**
YouTube Analytics API	Views, watch time, retention → daily learning and monetization progress	Free	**Connected**
Google OAuth	Sign in with Google per channel (official, no passwords stored)	Free	**Set**
Google Trends RSS	Today's most-searched stories in India (AI News)	Free, no key	**Working**
Google News RSS	News articles as evidence	Free, no key	**Working**
Wikipedia + pageviews	Background facts and topic demand	Free, no key	**Working**
Hacker News, RSS feeds, Ollama web search	Extra research sources	Free	**Working**
Meta Graph API	Post to Facebook Pages and Instagram	Free	**Not connected**

Pivot Reversal project files	NIFTY levels from your own app (no NSE scraping)	Local	**Used by NIFTY channel**

All keys are stored in the macOS Keychain (service `yt-automation-studio`) — never in the database, logs, Git or this document.

8. Your channels

#	Channel	Workflow	Language	Schedule
1	NIFTY 50 Fibonacci Levels	NIFTY daily education (compliance gate)	Hinglish	15:40 (off)
2	Kids and Comedy	Animated stories (original cartoon renderer)	Hinglish	18:00 daily
3	Stand-alone videos	Any request, download only	Hindi	Off
5	AI News	Trending news, AI presenter "Aarav Mehra", News category, synthetic disclosure	Hinglish	09:00 and 18:00 daily

9. Safety and rules built in

Facts
Every claim checked against sources; unsupported lines rewritten; still wrong → waits for you.

Copyright
No re-uploads of other people's videos; licensed stock with attribution; original music and art; copy detection on scripts.

AI disclosure
Realistic AI presenters are marked "altered or synthetic content" on YouTube.

Finance (SEBI)
No buy/sell or target wording; compliance gate and tamper-proof audit log for NIFTY videos.

Money
Free providers first, budget ₹0; paid ones skipped unless you allow them.

Reliability
Resume after crash, retries, IPv4 only, 5-minute limit on stuck AI calls, no duplicate tasks, wait instead of fail.

10. Current limits and next steps

Limit	Fix
Talking presenter only ~2 clips a day (free GPU minutes)	Fixed: SadTalker on the Mac makes the presenter talk in every video after LTX-2's minutes run out.
Free AI writers can all run out on a busy day	Videos now wait and retry. Optional: free Groq key, or a local Ollama model (~6 GB download).
Disk only ~12 GB free	Storage page: clear working files, or uninstall unused AI tools (local FLUX 6 GB, voice cloning 4.5 GB, Parler 1.4 GB).
Sarvam voice out of credits	Kokoro (free) is used instead.
Pexels and Meta not connected	Optional: add a free Pexels key; connect Facebook/Instagram when wanted.